

② 已知 $y''(t) + 4y'(t) + 5y(t) = f'(t)$, $y(0_-) = 1$, $y'(0_-) = 2$, $f(t) = e^{-2t}\varepsilon(t)$, 则 $y(t)$ 连续

$y(0_+) = \underline{1}$, $y'(0_+) = \underline{3}$

$y''(t) + 4y'(t) + 5y(t) = -2e^{-2t}\varepsilon(t) + e^{-2t}\delta(t)$ 只有 $y'(t)$ 有 $\delta(t)$ 项

$\int_{0_-}^{0_+} y''(t)dt + 4\int_{0_-}^{0_+} y'(t)dt + 5\int_{0_-}^{0_+} y(t)dt = \int_{0_-}^{0_+} -2e^{-2t}\varepsilon(t)dt + \int_{0_-}^{0_+} e^{-2t}\delta(t)dt$

(2) 已知描述系统的微分方程和初始状态, 写出其零输入响应。

$y(0_+) - y'(0_-) = 1$, $y(0_+) - y'(0_+) = 0$

① 已知 $y''(t) + 5y'(t) + 6y(t) = f(t)$, $y(0_-) = 1$, $y'(0_-) = -1$, 则 $y_{zi}(t) =$